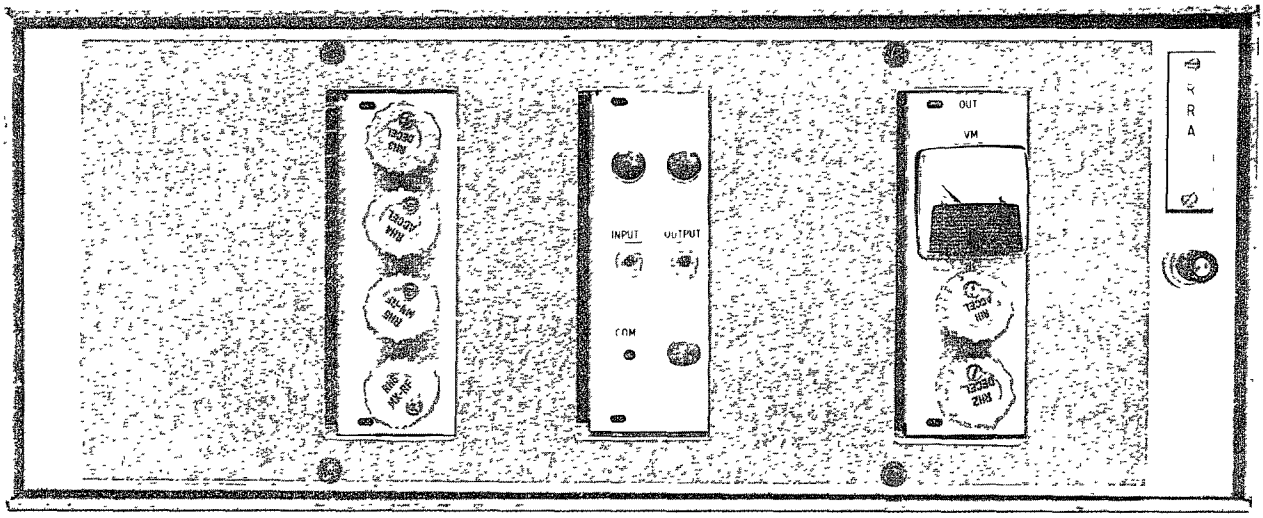
INSTRUCTIONS

RRA RATE REFERENCE AMPLIFIER



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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

RRA RATE REFERENCE AMPLIFIER

INTRODUCTION

The RRA rate reference amplifier provides a speed-reference voltage for speed-regulated sections of a paper mill drive or other equipment. This rate reference receives an input from a single reference source such as a speed-adjusting potentiometer. The output from the master rate reference is a stabilized d-c voltage. This output voltage is not sensitive to a-c line voltage fluctuations or temperature changes.

When used with a speed-adjusting potentiometer, a change in the potentiometer setting will cause the output of the rate reference to change to the new level at a predetermined rate-of-change. This rate-of-change is adjustable by means of acceleration or deceleration potentiometers on the front panel. These potentiometers provide for a variable rate-of-change to reach maximum output voltage from zero as follows:

	Time in Seconds
RRA	10 to 320

The output voltage from the rate reference is indicated on a 0-30 d-c voltmeter mounted on the front of the panel. This output is adjustable from zero to minus 30 volts d-c, and has a current rating of 20 milliamperes.

DESCRIPTION

All components of the rate reference, except the remotely located speed-adjusting potentiometer, are mounted inside an 8-inch high steel enclosure, commonly called a module. This 20-inch wide module is designed to mount on control panels where needed in a paper mill drive. Four mounting bolts which provide for convenient removal and replacement secure the module to the rack. All wires entering and leaving the module are connected to screw-type terminals accessible from the front when the cover plate is removed.

Those components visible from the front of the rate reference under operating conditions are shown in Fig. 1. When the front cover plate is removed, front access is obtained to the terminal boards

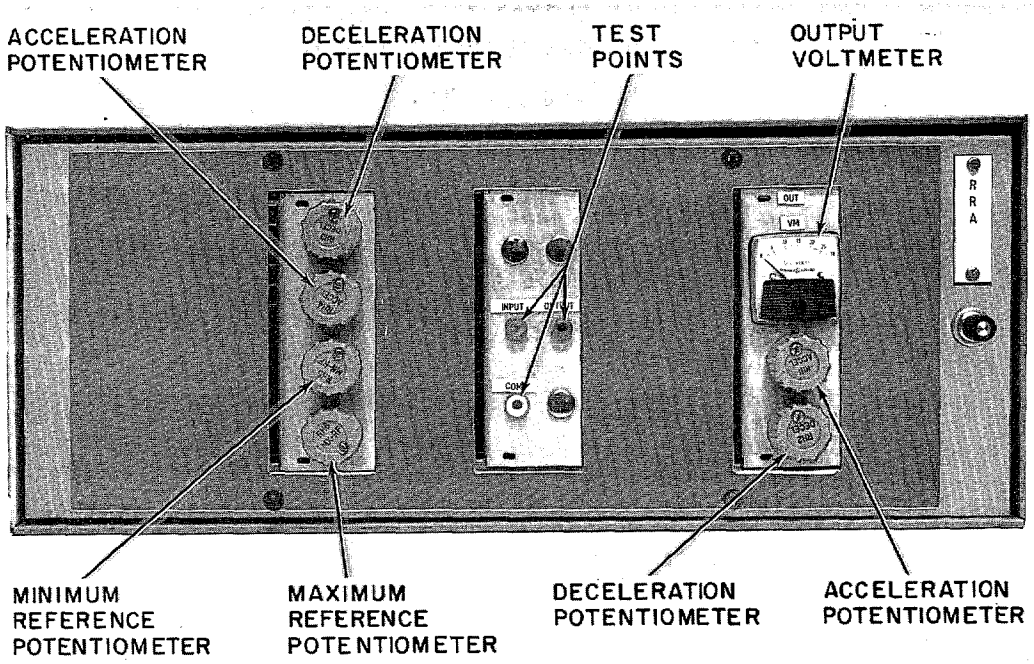


Fig. 1. Front view of RRA rate reference amplifier

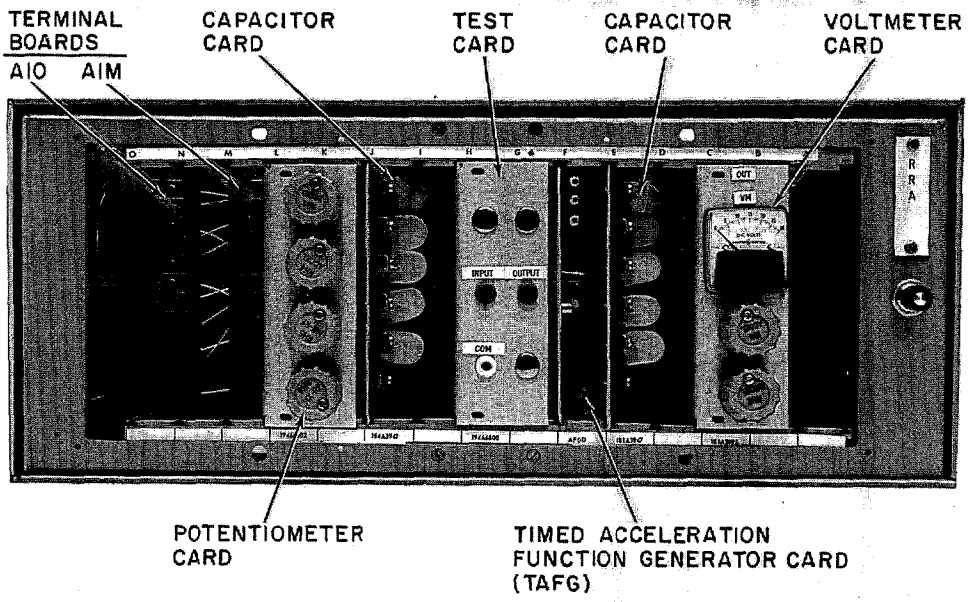


Fig. 2. Front view of RRA rate reference amplifier with cover removed

TABLE I
DEVICE NAMES AND LOCATIONS
(See Fig. 2)

SLOT DESIGNATION Alphabetical	CARD NAME	IDENTIFICATION NUMBER
O	Terminal Board TB1 (A1O)	None
N	Not Used	----
M	Terminal Board TB2 (A1M)	None
L	Pot Card	194A6802
K	Not Used	----
J	Capacitor Card	181A3947
I	Not Used	----
H	Test Card	194A6800
G	Not Used	----
F	Timed Acceleration Function Generator	IC3600AFGD1
E	Capacitor Card	181A3947
D	Not Used	----
C	Voltmeter Card	181A3995
B	Not Used	----
A	Not Used	----

and printed-wiring boards. These components are identified in Fig. 2 and Table 1.

A hinged door with latch provides access to all wiring within the enclosure as shown in Fig. 3. The printed-wiring board pins plug into a match receptacle mounted at the rear of the door assembly. All interconnecting wires between printed board receptacle pins are installed by means of a gastight pressure connection applied by a special tool. This tool simultaneously strips the insulation from the conductor, attached the lug, and compresses the lug and wire to the pin with a wipe-clean action. The conductor is held, under high pressure, against the surface of the connector pin to provide a gastight connection between conductor and pin.

The printed-wiring boards are made of glass epoxy to resist moisture absorption and warping in mill environments. After the board components are assembled and soldered, the entire board is conformal coated to protect it against dirt, moisture, fumes and other environmental contamination. When the boards are removed, a special card puller supplied with the equipment releases the printed board pins from the receptacle. During replacement, the back of this puller helps to seat the board pins into the mating receptacle. Both the board and receptacle contact pins are nickel-tin plated for good electrical contact and corrosion resistance.

OPERATION

GENERAL

A simplified diagram of the rate reference amplifier is shown in Fig. 4. All of the components except the speed-adjusting potentiometer are located in the module. The rate-of-change amplifier is a single printed-wiring board called the Timed Acceleration Function Generator (TAFG). This board (or card) is identified as IC3600AFGD1 and should not be confused with TAFG boards for other rate references that are similar in appearance.

An external power supply provides regulated positive and negative 50-volts d-c for all semiconductor circuits within the RRA rate reference amplifier.

RATE-OF-CHANGE AMPLIFIER

The rate-of-change amplifier on the TAFG board shown in Fig. 7 is a unity-gain voltage amplifier with an output voltage that responds to changes in input d-c voltage at a preset rate. This rate is controlled by an RC circuit and is adjusted by the acceleration and deceleration adjust potentiometers located on the front panel. Also provided within the module is a time-delay function on the rate-of-change. This function is controlled by the rounding capacitors. The rounding capacitors control the output voltage so that it moves smoothly into and out of the rate at the start and finish of an output voltage change. The magnitude of this rounding is adjustable within limits by jumpers on the module terminal board.

INPUT-OUTPUT CHARACTERISTICS

The output voltage of the rate reference is adjustable from zero to minus 30 volts; depending on the value of the input voltage to the rate-of-change amplifier. This output voltage is always negative with respect to common. The maximum output of

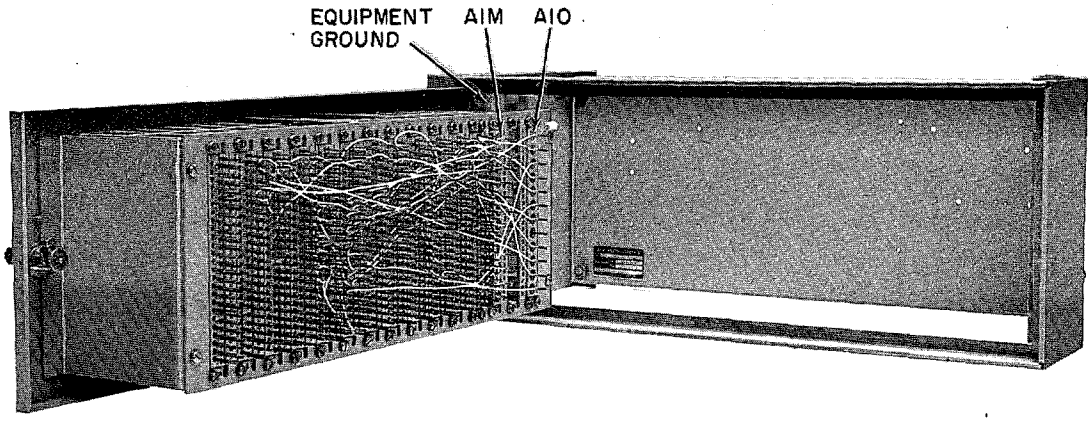


Fig. 3. RRA rate reference amplifier with door open showing interconnections

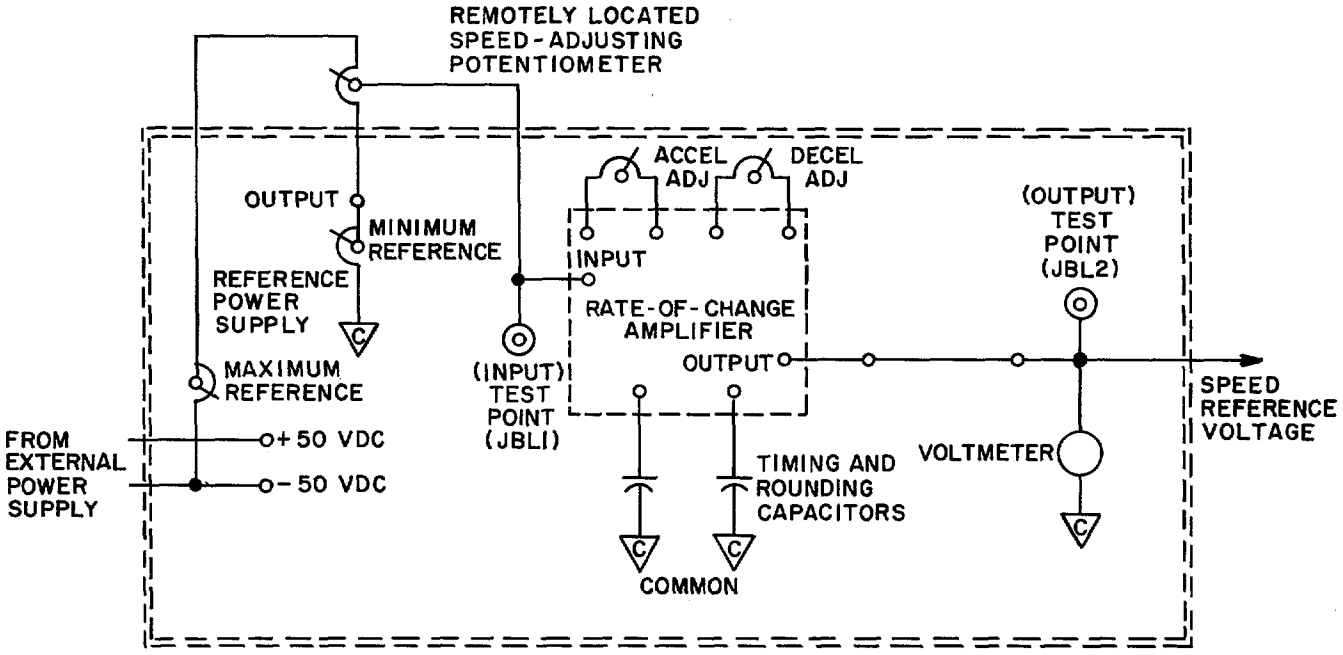
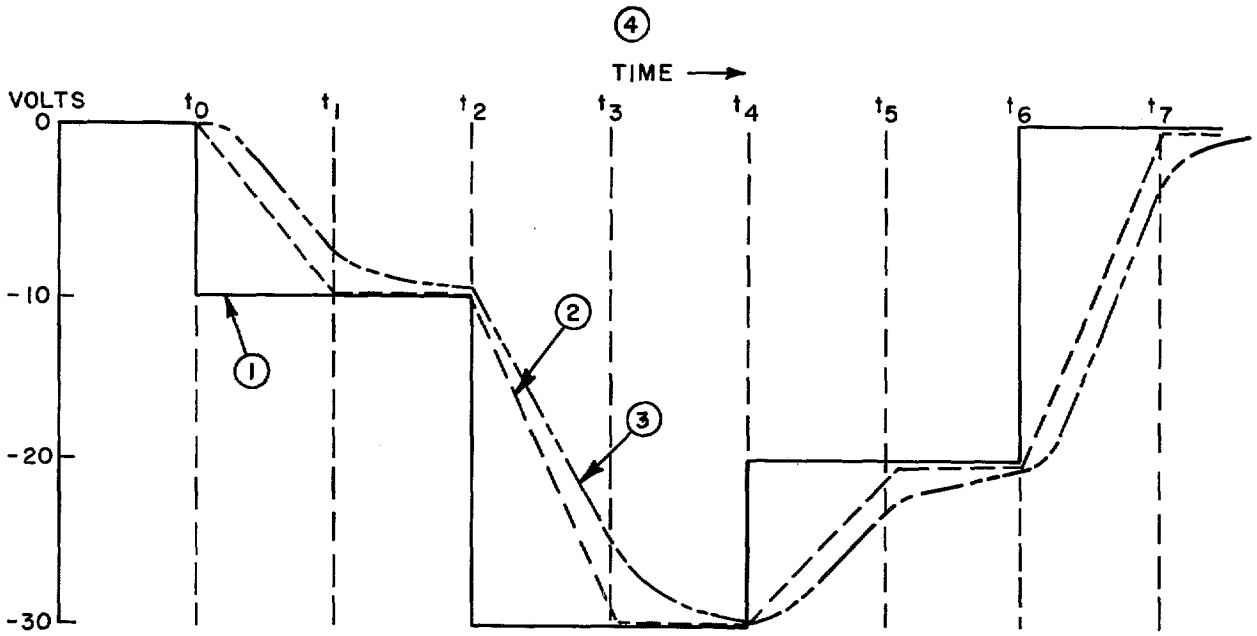


Fig. 4. Simplified diagram of RRA rate reference amplifier



① INPUT REFERENCE VOLTS - MEASURED AT JBL1

② OUTPUT VOLTS WITHOUT ROUNDING - MEASURED AT JBL2

③ OUTPUT VOLTS WITH ROUNDING - MEASURED AT JBL2

④ TIME INTERVALS SHOWN ARE NOT NECESSARILY EQUAL

Fig. 5. RRA input/output characteristics during acceleration and deceleration

RRA is 30 volts at 20 milliamperes. Both input and output curves of voltage vs. time are shown in Fig. 5. Curve No. 1 shows a change of input voltage to the rate-of-change amplifier to initiate a change in the speed reference. Curve No. 2 shows the rate output with no rounding; while Curve No. 3 shows the effect of the rounding on the rate output.

At time t_0, the input reference to the rate-of-change amplifier (Curve No. 1) is increased to minus 10 volts. Curve No. 2 shows that the output immediately changes at a constant rate of change until the output reaches minus 10 volts at t_1. The effect of rounding on Curve No. 3 shows the gradual change of the output before reaching the preset rate, and at t_1, the output begins changing gradually from its preset rate to the output set on the speed-adjusting potentiometer. Any increase in negative voltage at the output is defined as acceleration and this rate-of-change may be set by the acceleration potentiometer adjustment.

As shown on Fig. 5, at time t_2, the speed-adjusting potentiometer is changed to give a minus 30-volt input and the output changes to this new level reaching maximum output at t_4 with rounding. The rate-of-change remains the same; it merely takes longer to accomplish the 20-volt increase in output voltage.

At the time t_4, the input is reduced to minus 20 volts and the output starts decreasing at the predetermined constant rate. This decrease in negative voltage is defined as deceleration. It may be set independently of the acceleration rate by the deceleration potentiometer adjustment on the front panel. The input is reduced to zero volts at t_6 and deceleration continues at the predetermined rate until the output reaches zero.

VOLTMETER AND TEST POINTS

A voltmeter, VM, is provided on the output of the power amplifier. Test points are available at the input and output of the rate-of-change amplifier. These test points are identified as JBL1 and JBL2 respectively. All voltage measurements taken at these test points are made with respect to common.

INSTALLATION

No special installation instructions are required since the RRA rate reference is normally mounted and wired at the factory. When mounted in the field, position the module for unrestricted air flow at a height convenient for access to the controls. Allow sufficient slack in the wiring to the terminal boards for free door swing.

INSTALLATION ADJUSTMENT

Since the internal trimpots on the TAFG card are set and sealed at the factory, they should not require adjustment in the field. If system tune-up indicates the need for a realignment of these potentiometers it is recommended that the card be replaced. Refer to Appendix A of this publication for additional information concerning theory and adjustments of the potentiometers on the TAFG card.

The following adjustments are those required to set the rate reference and rounding characteristics for desired mill operation. Reference to the RRA Elementary Diagram (Fig. 7) will be helpful when performing these adjustments.

RATE ADJUSTMENT

Both the acceleration and deceleration rates for the mill drive are set independently by the "accel" (RH2) and "decel" (RH1) adjustment potentiometers shown in Fig. 1 and elementary diagram, Fig. 7. These potentiometers (calibrated from 0 to 10) set the acceleration and deceleration rates within the following range:

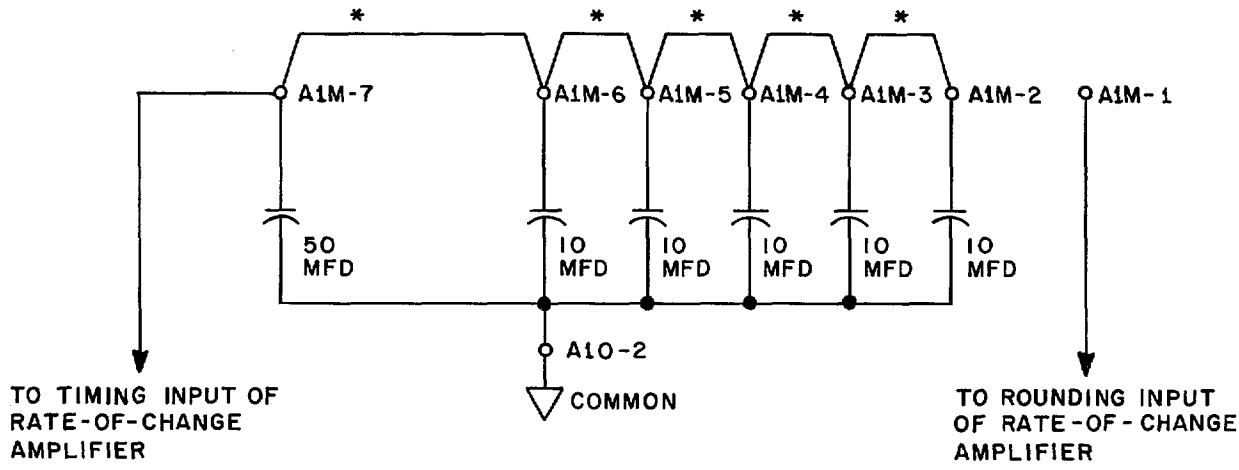
Rate Reference	Timing Range to Maximum Speed
RRA	10-320 seconds

This range is approximate and may vary ± 10 percent. This is the time required to obtain maximum speed from zero when the speed-adjusting potentiometer calls for 100-percent speed.

In addition to the "accel" and "decel" potentiometers RH2 and RH1, two other potentiometers RH3 and RH4 are used to recalibrate the timing of the rate reference using external interlocks to parallel the normal "accel" and "decel" potentiometers when recalibration is required. When recalibration is used, the parallel resistance of RH2 and RH4 or RH1 and RH3 sets the "accel" and "decel" time.

RATE AND ROUNDING CAPACITORS

The maximum acceleration and deceleration times are established by the amount of capacitance connected between A1M-7 to common and from A1M-1 to common as shown in Fig. 6. For the RRA rate reference, the total internal capacitance is 100 mfd, consisting of multiples of 10-mfd capacitor units. Jumpers on terminal board A1M provide for changing the distribution of this capacitance between the timing and rounding circuits at tune-up



* REMOVABLE EXTERNAL JUMPERS ON TERMINAL BOARD (A1M)

Fig. 6. Terminal connections for rate and rounding capacitors

for desired mill-drive operation. After factory tests, 100-mfd capacitance is connected to the timing input of the rate-of-change amplifier (A1M-7 to common). If necessary, external capacitance may be connected from A1M-7 to common and A1M-1 to common (A1O-2).

a. If no rounding is required, the total 100-mfd capacitance is applied to the timing input. The corresponding results are shown in Table 2.

b. When rounding is required, the capacitors should be reconnected as shown in Table 2. For slight or maximum rounding effects, transfer capacitors between A1M-7 and A1M-1 by means of jumpers. The results are shown in Table 2.

TABLE II

RRA TIMING EFFECTS WITH AND WITHOUT ROUNDING

Setting of "Accel." Adjust (See Note 2)	No Rounding			Slight Rounding			Maximum Rounding		
	Accel. Time (Sec)	A1M-1 (Mfd)	A1M-7 (Mfd)	Accel. Time (Sec)	A1M-1 (Mfd)	A1M-7 (Mfd)	Accel. Time (Sec)	A1M-1 (Mfd)	A1M-7 (Mfd)
0.0	11	0	100	14	10	90	21	30	70
0.7	32	0	100	35	10	90	44	40	60
3.0	104	0	100	108	10	90	120	50	50
10.0	316	0	100	317	10	90	330	50	50

NOTE 1: The above values may vary plus or minus 10 percent due to component tolerances.

NOTE 2: The deceleration time from maximum speed to zero will be the same as the accelerating time for equal settings of the "accel" and "decel" adjust potentiometers.

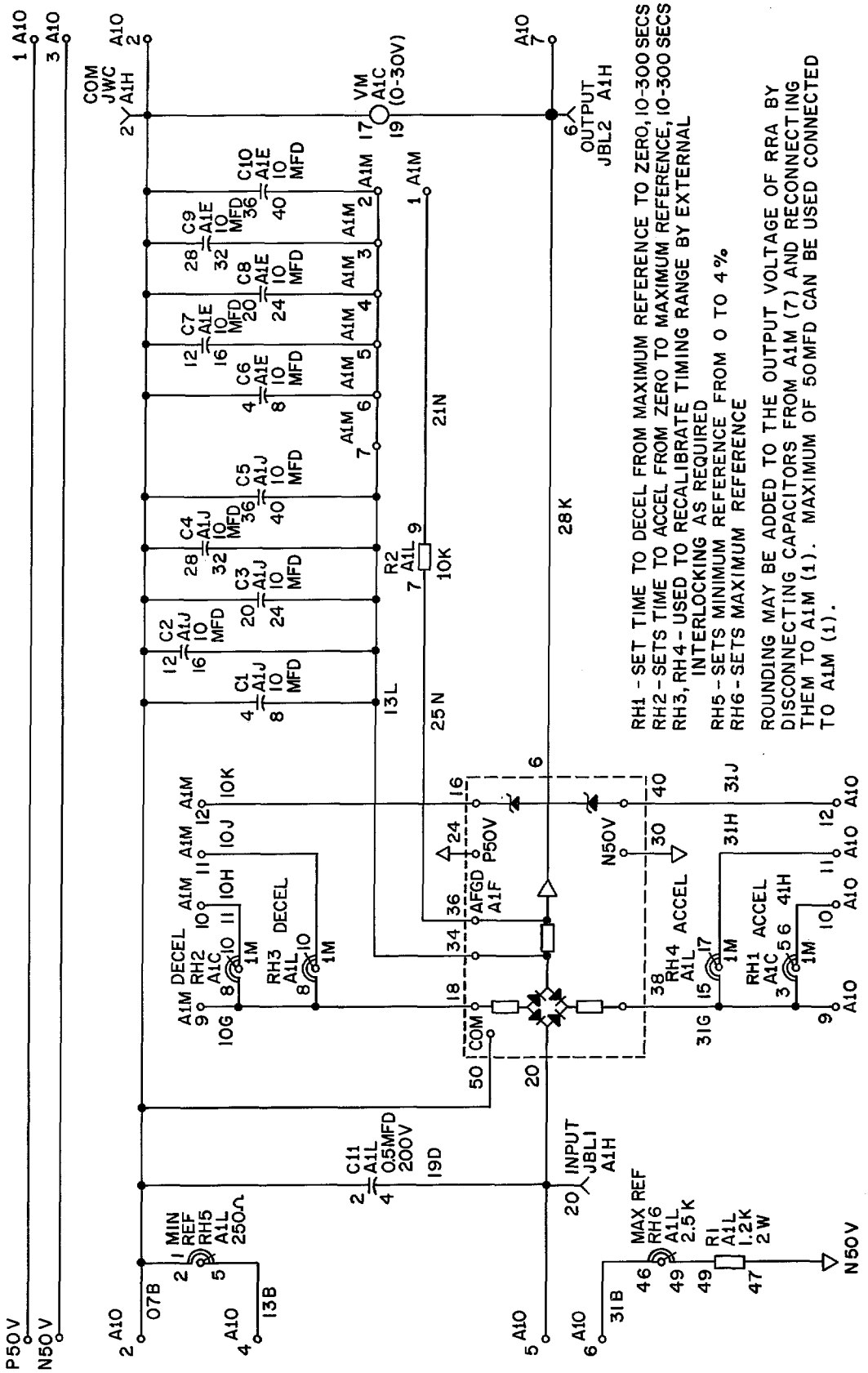


Fig. 7. RRA rate reference amplifier, elementary diagram. This diagram is typical. Always refer to elementary diagram furnished with equipment

TABLE III
TROUBLESHOOTING CHART

Symptom	Procedure
1. No speed reference voltage on output voltmeter, VM.	1) Check for plus 50 volts on A1O-1 and -50 volts on A1O-3 with respect to common (A1O-2). If only one or neither voltage is present, check output of the external power supply. 2) Disconnect panel wire on A1O-7. If VM then shows a voltage with speed reference input to the module available, check for a short on the reference output wiring. 3) Check input at test point JBL1. Vary input reference to see that input is not shorted to common. If no input voltage is available and input is not shorted to common, check A1O-6. This should be about -30 volts supplying the speed reference circuit. If -30 volts is not available, check maximum reference potentiometer RH6.
2. Output voltage at maximum; no control with input voltage changes.	1) Check plus or minus 50-volts from external power supply. 2) Check input voltage at JBL1. With an open input circuit (zero volts), the output will drift up to a maximum. If the input circuit is not open, measure JBL1 and JBL2. These voltages should differ by not more than 100 millivolts. If they differ by more than this, replace TAFG card.
3. Speed reference will accelerate but not decelerate.	1) Check "decel" adjust potentiometer for continuity. Replace if open. If not open, replace TAFG card.
4. Speed reference will decelerate but not accelerate.	1) Check "accel" adjust potentiometer for continuity. Replace if open. If not open, replace TAFG card.
5. Only maximum accelerating and/or decelerating times available. No rate adjustment.	1) Check "accel" and/or "decel" adjust potentiometers for open slider contact. Replace if open.
6. Master Speed Reference output follows master speed potentiometer instantaneously.	1) Check to see that timing capacitors are connected. 2) Replace TAFG card.

Table 2 shows the acceleration time for RRA from zero to maximum speed for four settings of the "accel" adjust potentiometer. This table also shows the effects of slight of maximum rounding.

MAINTENANCE

INSPECTION

At regular intervals, or while performing maintenance in other areas near the RRA rate reference, visually inspect the module for:

1. Loose terminals at A1M and A1O.

2. Frayed or damaged insulation on all exposed wiring.

3. Loose mounting screws at receptacles or power supply heat sinks.

4. Printed wiring boards not fully inserted in receptacles.

5. Loose wire lugs on receptacle pins. Solder wire and lug directly to receptacle pin if special application tool* is not available. Use good quality resin-core solder and small iron or soldering gun to apply heat.

TROUBLESHOOTING

In the event of trouble within the rate reference, the procedure listed in the TROUBLESHOOTING table is recommended. Voltages should be measured with a volt-ohm-milliammeter having a sensitivity of 20,000 ohms per volt or better. Typical voltage levels at the test points during normal operation are summarized in Table 4.

*A maintenance kit, Catalog No. IC3603A131A is available. Consult your General Electric Sales Representative.

TABLE IV

RRA OPERATING RANGES

(With Respect to Common, A10-2)

Function	Test Point	Time or Voltage (DC)	Remarks
Positive Bus Voltage	A1O-1	+50, ±1 volt	
Negative Bus Voltage	A1O-3	-50, ±1 volt	
Input	JBL1	0 to -30 volts	Depending on Speed Pot Setting
Output	JBL2	0 to -30 volts	Should follow JBL1 within ±100 MV
Accel Adjust Pot	JBL2	10 to 320 sec*	
Decel Adjust Pot	JBL2	10 to 320 sec*	

*Time to reach maximum output from zero with maximum input.

APPENDIX A
TIMED ACCELERATION FUNCTION GENERATOR CARD (IC3600AFGD1)

CHARACTERISTICS

1. Input Power	±50 volts, 10 milliamperes - 50 volts, 50 milliamperes ± load current
2. Load	
A. Rate circuit (Pin 6)	0 to -30 volts, 0 to 20 milliamperes
3. Output Ripple	Less than 10 millivolts peak-to-peak
4. Temperature Drift	
A. Rate circuit voltage drift (Pin 6)	0.004 percent of rated output/°C
b. Rate circuit current drift (Pin 20)	±0.01 microamperes/°C
5. Input Impedance (Pin 20)	Greater than 25 megohms d-c
6. Output Impedance	
A. Rate circuit (Pin 6)	Less than 2 ohms
7. Timing	Capacitor charging current adjustable by external resistance from 12 to 285 microamperes

FUNCTION

The rate reference, or more accurately, the rate-of-change-limited precision voltage reference, is a unity-voltage-gain amplifier (Fig. A6) with an output voltage that responds to changes of input d-c voltage at a preset rate. It also possesses an "S Curve" characteristic whereby the output voltage moves smoothly into and out of this rate at the start and finish of the voltage change.

THEORY OF OPERATION

The rate-of-change of voltage is generated at the immediate input to the amplifier by the charge and discharge of a timing capacitor from current sources I_D and I_C. Figure A1 shows the steady-state condition.

When V_{in} is changed rapidly to a different level, a diode bridge effectively breaks the input circuit connections as shown in Fig. A2 and A3. At the point when V_c reaches V_{in}, the circuit connections revert to those in Fig. A1.

The rate-of-change of capacitor voltage, V_c, is determined by the current sources and the capacity.

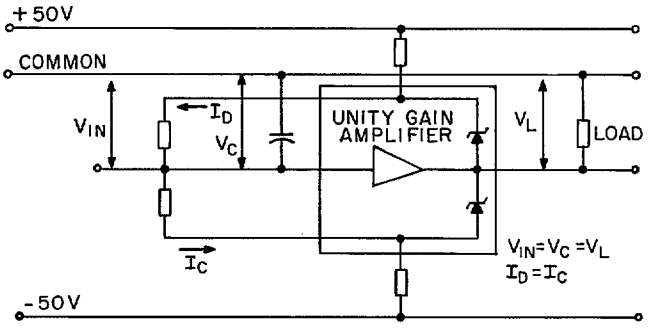


Fig. A1. Steady-state condition

The magnitude of each current source is related to the internal zener voltages (10 volts), the forward drop of a silicon diode, and the timing resistors.

$$\frac{dVc}{dT} = \frac{I}{C} = \frac{10. - 0.5}{RC} \text{ Volts/Second}$$ (R - Megohms, C - Microfarads)

An "S Curve" characteristic is obtained by replacing the single capacitor with a Pi-connected R, C1, C2 combination as shown in Fig. A4. The

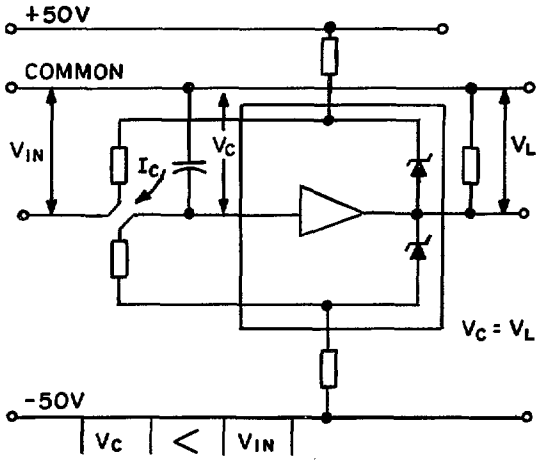


Fig. A2. Charging conditions

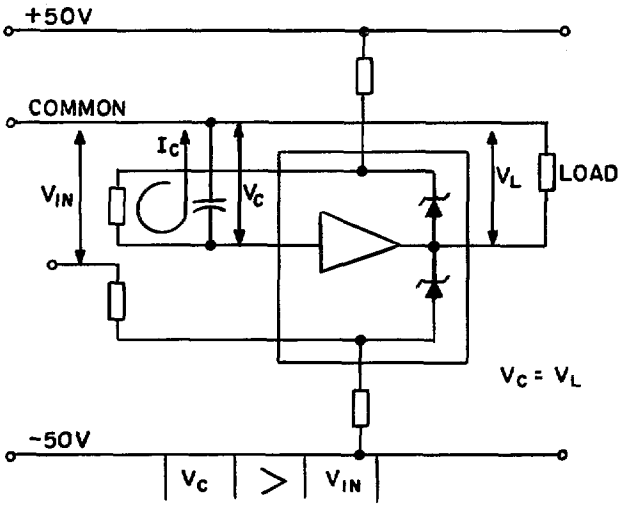


Fig. A3. Discharging conditions

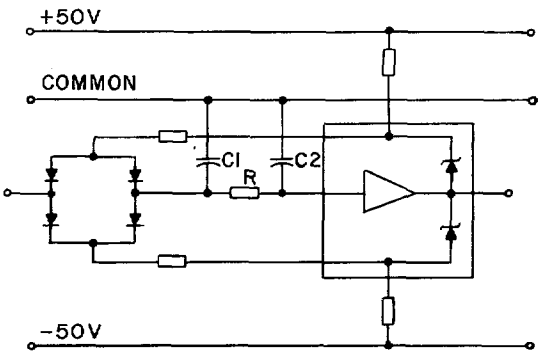


Fig. A4. Basic circuit

components R and C2 form a lag circuit for both increasing and decreasing input voltages. Also shown in Fig. A4 is the diode bridge.

CARD ADJUSTMENTS

Three trimpots (Fig. A5) are provided on the rate reference; an output zero adjustment, R44; a d-c bias control, R42; and a trimpot, R43, that prevents the output from drifting positive with respect to common.

The effects of the controls on the output voltage are interrelated. All trimpots are set in test and normally do not require adjustment in the field. If adjustments are required they should be made as follows:

1. Zero Adjust R44 – Connect 1-megohm 1-percent resistors across Pins 16-18 and Pins 38-40. With the clamp disabled and with zero bias current (R42 CCW and R43 fully clockwise) and input shorted to common, adjust R44 for zero output voltage.

2. Bias Adjust R42 – Connect a 10-mfd capacitor across Pin 34 and common. Switch input to 30 volts negative. Time the change of output voltage up to 30 and back to zero. Advance R42 clockwise until the "UP" and "DOWN" times are equalized. Readjust R44 for zero output voltage.

3. Clamp Adjust R43 – Remove the 10-mfd capacitor. With input shorted to common, advance the output voltage to 5 millivolts negative with respect to common. Check that output will not go positive

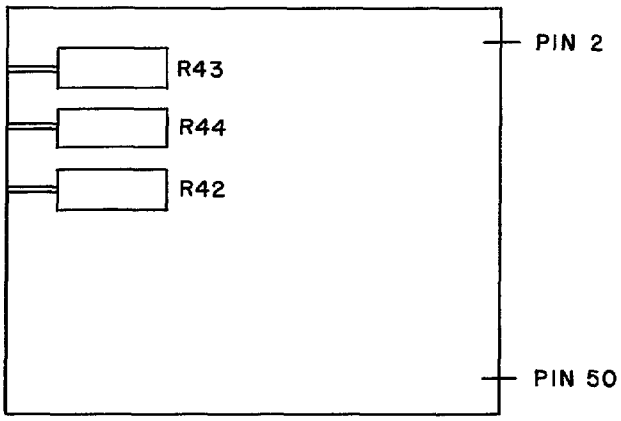


Fig. A5. Location of trimpots on TAFG card

by swinging R44. Return R44 to setting giving the negative 5 millivolts output voltage.

SELECTION OF TIMING CAPACITORS AND RESISTORS

Both C1 and C2 determine the rate-of-change of output voltage; however, only C2 controls the shape of the "S Curve." When C2 is added, the rounding at the start and end of acceleration is not equal. The rounding at the end of acceleration may be very long. To minimize this effect, it is recommended that the product of C2 in microfarads and the acceleration in volts/second should not exceed 15.

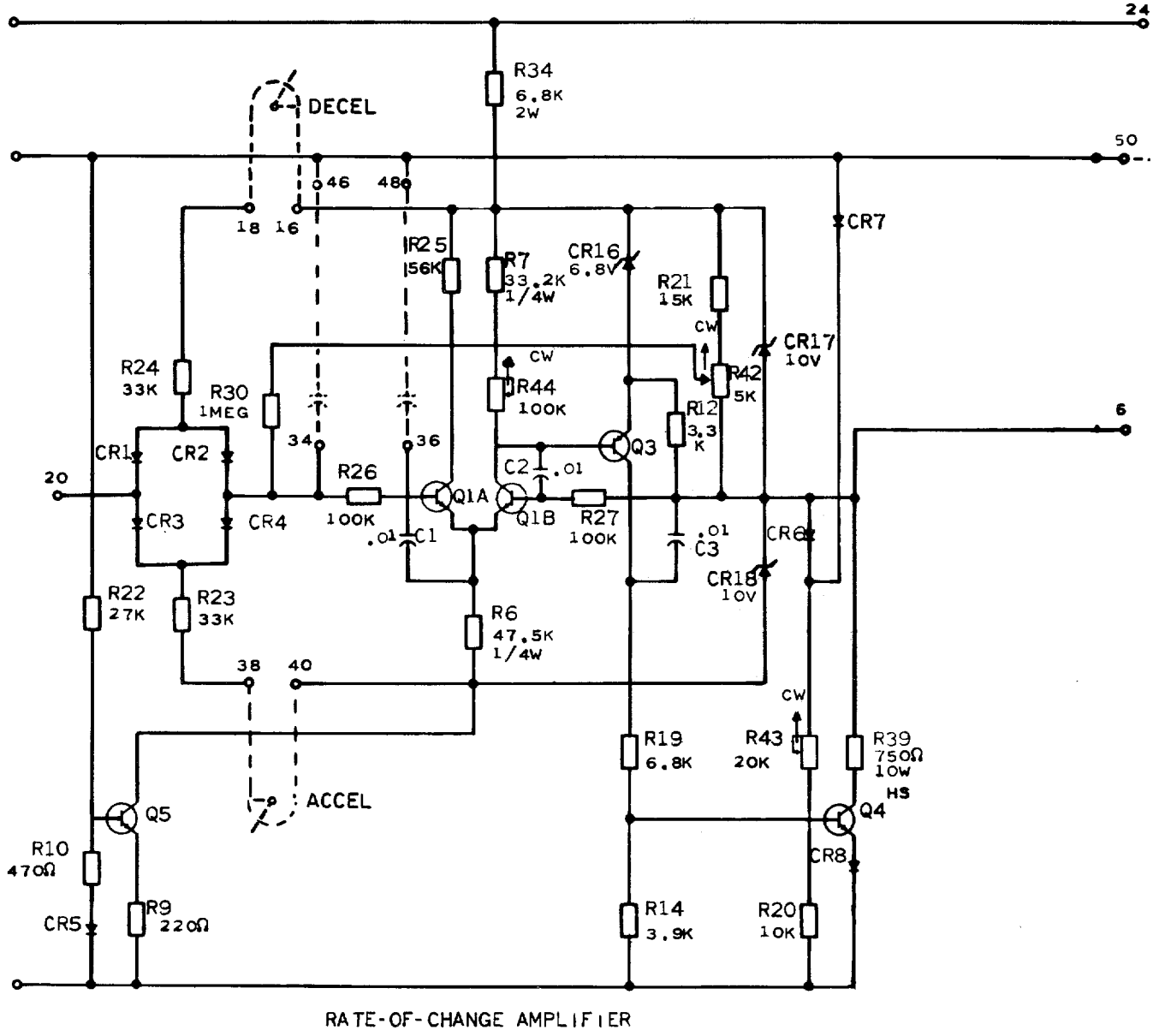


Fig. A6. Elementary diagram of timed acceleration function generator card

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